MechMind AI

General Predictive Maintenance & Troubleshooting Guide

Expanded Edition: condition indicators, safe troubleshooting workflows, ML score interpretation, dataset notes, checklists, scenarios, and RAG knowledge-base governance

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Important scope note
This is a general educational guide, not an OEM service manual. It does not provide machine-specific limits, torque values, lubrication grades, repair steps, or lockout procedures. Always follow the machine manufacturer's documentation and your site's approved safety procedures.

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1. Purpose, scope, and how to use this guide

1.1 Purpose

Predictive maintenance uses condition data and operating history to help identify changes that may indicate developing equipment problems. It complements, rather than replaces, scheduled maintenance, operator inspections, and manufacturer recommendations. The goal is earlier, better-informed decisions: catching a developing problem while there is still time to plan the response, and avoiding unnecessary interventions when readings are simply normal variation.

This guide serves two audiences at once. People can read it as a primer on indicators, workflows, and record keeping. The MechMind AI assistant retrieves passages from it to answer questions, so each section is written to stand on its own, with clear headings, definitions, and explicit statements of what the guidance does and does not cover.

1.2 What this guide is and is not

This guide IS	This guide is NOT
A general explanation of common condition indicators and what they may suggest.	An OEM service manual or a substitute for one.
A structured, safety-first troubleshooting approach that works across equipment types.	A source of machine-specific limits, torque values, clearances, or lubricant grades.
A guide to interpreting machine-learning failure scores with appropriate caution.	A validated predictive model. The current model is trained on synthetic data.
A set of checklists, templates, and scenarios you can adapt to your site.	A lockout/tagout procedure or authorization to perform repairs.
Governance advice for keeping a RAG knowledge base accurate and citable.	Legal, regulatory, or insurance advice.

1.3 Fields and data referenced

The guide focuses on general industrial equipment and on the sensor fields represented in the AI4I 2020 predictive-maintenance dataset: air temperature, process temperature, rotational speed, torque, tool wear, and machine type (product quality variant). The dataset is synthetic; patterns learned from it should not be treated as validated thresholds for a real machine. Section 7 describes the dataset in detail, and Section 8 explains how to read the model's output.

1.4 Order of precedence when instructions conflict

If this guide ever appears to conflict with another source, follow this order, highest priority first:

- Applicable laws, regulations, and your site's emergency and safety procedures.
- The machine manufacturer's (OEM) documentation for the specific equipment and revision.
- Site-approved maintenance procedures and work instructions.
- Qualified technician or engineer judgment applied to the actual situation.
- This general guide, which is background knowledge only.

2. Maintenance strategies and where predictive maintenance fits

Predictive maintenance is one of several strategies. Most real programs combine them, choosing per asset based on criticality, failure consequences, cost, and how detectable the failure is in advance.

Strategy	How it works	Strengths	Limitations
Reactive (run-to-failure)	Act after something fails.	No monitoring cost; suits low-criticality, cheap, redundant items.	Unplanned downtime, secondary damage, and possible safety exposure.
Preventive (time or usage based)	Service at fixed calendar or usage intervals.	Predictable planning; simple to schedule.	May replace healthy parts or miss failures that develop between intervals.
Condition-based	Act when a measured condition crosses an approved trigger.	Interventions follow actual need.	Needs reliable sensors and justified triggers from OEM or site engineering.
Predictive	Use trends and models to estimate developing problems and remaining useful life.	Earlier warning; better planning.	Needs quality data, validation, and human review; models can be wrong.
Reliability-centered maintenance (RCM)	Choose tasks per failure mode based on consequences.	Aligns effort with risk; documents reasoning.	Analysis effort up front; needs periodic review.

2.1 The P-F interval idea

A developing failure usually passes through a point where a change first becomes detectable (potential failure, P) and a later point where the equipment can no longer do its function (functional failure, F). The time between P and F is the P-F interval. Monitoring only helps if the checks happen more often than the P-F interval is long, and if the response (inspect, plan, repair) fits inside the remaining time. Different indicators have different P-F intervals, which is one reason to review several indicators together.

2.2 Where an ML model fits

A model that estimates failure risk is a decision-support layer on top of the strategies above. It can help prioritize which machines to look at first. It does not replace operator observation, OEM inspection intervals, or safety procedures, and it should never be the only reason to continue running equipment that shows signs of a hazard.

3. Common condition indicators

Each indicator below is described the same way: what it is, why it matters, factors that may contribute to changes, what to review, and common pitfalls. The listed factors are possibilities to investigate, not diagnoses. A single value rarely proves a root cause.

3.1 Quick reference

Indicator	What to review	Possible follow-up
Temperature	Compare readings with the machine's normal baseline and approved operating range.	Check sensor quality, cooling/ventilation, load changes, and operating conditions.
Rotational speed	Look for unexpected changes or unstable readings relative to operating mode.	Confirm readings and setpoints; review drive or control alarms using the OEM guide.
Torque / load	Compare load readings across similar operating conditions.	Review process load, jams, friction, and sensor validity under approved procedures.
Tool wear	Track accumulated usage against approved inspection or replacement schedules.	Inspect tooling using the manufacturer's method and authorized procedures.
Vibration / noise	Note new or increasing vibration, unusual noise, or changes from baseline.	Use approved monitoring methods; escalate if safety or integrity is in doubt.

3.2 Temperature (air temperature and process temperature)

What it is. Air (ambient) temperature describes the surroundings. Process temperature describes the temperature of the process, component, or working zone as measured by the machine's sensor. The relationship between the two, often the difference, can be as informative as either value alone.

Why it matters. Heat is a by-product of friction, electrical losses, and process load. Many degradation mechanisms (lubricant breakdown, seal aging, insulation stress, thermal expansion) are temperature-sensitive, so a sustained change from baseline can be an early hint worth reviewing.

Factors that may contribute to change.

- Ambient changes: season, time of day, nearby heat sources, enclosure ventilation.
- Reduced heat removal: blocked airflow, dirty filters or fins, cooling or fan problems, coolant issues.
- Higher mechanical or process load; increased friction; lubrication condition (check OEM guidance).
- Operating-mode changes such as warm-up, idle, or heavy-duty cycles.
- Sensor problems: drift, poor contact, wiring faults, wrong units, or incorrect placement.

What to review. Trend rather than single points; the same operating mode; ambient temperature at the time; the temperature difference between process and air; and whether other indicators (torque, speed, vibration) moved at the same time.

> **Pitfall**
> Check units before anything else. Datasets and controllers may report Kelvin, Celsius, or Fahrenheit. A unit mismatch can make a normal machine look alarming or hide a real problem. A rapid rise, smoke, or a burning smell is an immediate-hazard sign: follow site emergency procedures.

3.3 Rotational speed

What it is. The measured rotational speed of a spindle, shaft, or drive, usually in revolutions per minute (rpm).

Why it matters. Speed reflects both what the control is asking for (setpoint) and what the mechanics can deliver under load. A gap between setpoint and actual speed, or unstable speed, can indicate overload, slip, or control problems.

Factors that may contribute to change.

- Setpoint or program changes; different operating mode or product.
- Increased load pulling speed down; mechanical drag or binding.
- Belt or coupling slip, where applicable.
- Drive, control, or power-supply issues; active alarms.
- Speed sensor or encoder faults, noisy signals, or scaling errors.

What to review. Actual versus setpoint; stability over time (hunting or oscillation); speed behaviour at the moment torque or temperature changed; and drive/control alarm history interpreted using the OEM guide.

3.4 Torque and load

What it is. Torque is the rotational force at the shaft or tool, commonly in newton-metres (Nm). Load indicators such as motor current or power can serve a similar purpose when direct torque is not measured.

Why it matters. Torque reflects how hard the machine is working. Rising torque for the same task can suggest increased resistance (for example worn tooling, friction, or misalignment). Sudden spikes may indicate jams or process disturbances. Unusually low torque can also be informative, for example when the tool is not engaging.

Factors that may contribute to change.

- Material or product variation, depth or feed changes, process parameter changes.
- Tool condition: dull, damaged, or incorrectly installed tooling.
- Binding, misalignment, or lubrication-related friction (follow OEM guidance).
- Jams, debris, or obstruction in the process.
- Sensor calibration issues or electrical noise in the signal.

What to review. Compare torque only across similar operating conditions. Look at both averages and peaks, and note whether changes were gradual (more typical of wear or fouling) or abrupt (more typical of jams or events).

> **Useful relationship**
> Relationship worth knowing: mechanical power is torque multiplied by angular speed (angular speed in radians per second = rpm x 2 x pi / 60). If speed falls while torque rises, the product may stay similar even though the machine is under more strain. Always look at both.

3.5 Tool wear

What it is. An accumulated measure of tool usage, often in minutes of use or cycles, used to schedule inspection or replacement. In the AI4I 2020 dataset it is expressed in minutes.

Why it matters. Worn tools can raise torque and temperature, degrade output quality, and in some cases fail suddenly. Usage-based counters are convenient but approximate, because actual wear depends on material, speed, feed, coolant, and tool quality.

What to review. The counter against the approved inspection or replacement schedule; output-quality trends (surface finish, dimensional drift); and secondary signs such as rising torque or temperature at similar settings. Inspect tools using the manufacturer's method and only through authorized procedures.

Pitfall
A tool-wear counter is only as good as its reset discipline. If a tool is changed and the counter is not reset (or the reverse), the data will mislead both people and models. Record every tool change.

3.6 Vibration

What it is. Oscillatory motion of a machine or component, characterised by amplitude, frequency content, and trend over time.

Why it matters. Rotating machinery produces a baseline vibration signature. Changes from that signature can accompany imbalance, misalignment, looseness, bearing wear, or resonance. Vibration analysis often gives some of the earliest indications of mechanical change.

What to review. Trend versus baseline measured at the same location, direction, and operating condition; whether the change is broadband or concentrated at particular frequencies; and whether temperature or noise changed too. Meaningful interpretation requires proper sensors and trained analysts using approved methods, and acceptance limits must come from the OEM or applicable standards, not from generic numbers.

3.7 Noise and acoustic observations

Operators often notice new sounds before instruments do: grinding, squealing, knocking, rattling, or a change in pitch. Record what was heard, where, when (start-up, steady state, load changes), and for how long. Treat sudden loud noise, or noise accompanied by smell, smoke, or heat, as a potential immediate hazard.

3.8 Electrical load: current and power

Motor current and electrical power can serve as a proxy for mechanical load when torque is not measured directly. Compare across the same operating mode. Changes may accompany mechanical drag, process load changes, or electrical issues. Electrical inspection and measurement should only be done by qualified, authorized personnel.

3.9 Lubrication, fluids, and contamination

Lubricant level, condition, and cleanliness influence friction and heat. Fluid analysis (when part of the site program) can reveal contamination or wear particles. This guide does not specify lubricant grades, quantities, or intervals; use only OEM-approved lubricants and procedures.

3.10 Pressure and flow (hydraulic, pneumatic, coolant)

Where present, pressure and flow readings help show whether supporting systems are delivering what the process needs. Pressure loss, pulsation, or unusual temperature in these systems may be related to leaks, blockages, or pump or valve condition. Pressurised systems store energy and must be handled under site-approved isolation procedures.

3.11 Operator observations and context

Structured operator notes (unusual smell, visible leaks, changed output quality, recent changeovers, material batches, weather, power events) frequently explain sensor patterns and provide labels that improve later analysis. Encourage short, consistent notes using the record checklist in Section 10.

4. Reading several indicators together

Single indicators are easy to misread. Patterns across indicators narrow the list of hypotheses. The table below lists common patterns with *possible* interpretations and sensible next reviews. These are starting points for investigation by qualified people, not conclusions.

Pattern observed	Possible interpretations to investigate	Suggested next review
Temperature up, torque up, speed stable	Higher process load, increased friction, tool condition, reduced cooling.	Compare with load history; check tooling and cooling per OEM and authorized procedures.
Temperature up, torque and speed normal	Ambient change, cooling or ventilation issue, sensor drift or placement.	Check ambient data, airflow, sensor validity against a second reading.
Speed down, torque up	Overload, jam, slip, or mechanical drag.	Review process conditions and drive alarms; escalate if load is severe.
Speed unstable, torque normal	Control/drive tuning, encoder or sensor signal issues.	Review control alarms and signal quality using OEM documentation.
Torque spikes at specific points in the cycle	Process disturbance, material inclusions, tool engagement, mechanical interference.	Correlate spikes with cycle phase; review recent process or tooling changes.
High tool wear, rising torque, worse output quality	Tool nearing end of useful life.	Inspect tooling per manufacturer method; verify the wear counter was reset at the last change.
Vibration up with temperature up	Bearing, alignment, or looseness concerns.	Escalate for qualified vibration analysis; consider risk of continued operation.
Several channels flatline or jump at the same instant	Data acquisition or communication fault rather than a real change.	Check logger, network, power, and timestamps before acting on the readings.
Model score rises, sensors look ordinary	Model sensitivity, subtle combined shift, concept drift, or a false alarm.	Review the feature values behind the score; compare with a valid baseline; ask for an operator check.
Model says no failure, operator reports unusual noise	Model blind spot (no acoustic input), or early issue not captured in the training data.	Trust the observation enough to inspect safely; record the case for future model review.

5. A practical troubleshooting workflow

Use the same sequence every time. It keeps safety first, avoids jumping to conclusions, and produces records that make the next investigation easier.

5.1 Step by step

- **Step 0: Safety triage.** Before analysing data, ask whether anyone is at risk. Smoke, burning smell, rapid temperature rise, severe vibration, fluid release, exposed energized parts, or guard/interlock problems mean: follow site emergency procedures and involve qualified personnel immediately.
- **Step 1: Confirm the observation.** Record time, machine identifier, operating mode, sensor values, alarms, and recent changes. Confirm sensors report plausible values (Section 6).
- **Step 2: Gather context.** Recent maintenance, tool changes, product or material changes, environmental changes, power events, previous similar events, and operator comments.
- **Step 3: Compare with a valid baseline.** Compare readings only with a baseline for comparable operating conditions. Do not treat generic values as manufacturer limits.
- **Step 4: Review multiple indicators.** Consider temperature, speed, load/torque, wear, vibration, alarms, and maintenance history together (Section 4). A single value rarely proves a root cause.
- **Step 5: List hypotheses.** Write down two or three plausible explanations and what evidence would support or rule out each.
- **Step 6: Start with the least invasive checks.** Prefer data review, remote checks, and visual observation from a safe position before anything that requires access to hazardous areas.
- **Step 7: Assess risk and decide.** Continue monitoring, schedule inspection, reduce load per approved procedures, or stop the machine. Site policy and qualified personnel make this call.
- **Step 8: Inspect or repair only when safe and authorized.** Follow site-approved isolation and lockout/tagout procedures before physical inspection or repair. Never bypass guards or interlocks.
- **Step 9: Verify.** After authorized work, verify operation using approved procedures and continue monitoring for a defined period.
- **Step 10: Document and learn.** Record findings, actions, parts used, and outcome. Feed confirmed outcomes back into baselines and model evaluation.

5.2 Example severity levels (adapt to your site)

Many sites classify condition findings so that responses are consistent. The levels below are an illustration; actual definitions and response times belong in your site's procedures.

Level	Typical meaning	Typical response
Information	Small change, within normal variation or explained by a known cause.	Log it; continue routine monitoring.
Watch	Persistent or growing change from baseline, no immediate hazard.	Increase monitoring frequency; plan an inspection.
Warning	Clear deviation, multiple indicators agree, or the model score is high and confirmed by review.	Schedule an authorized inspection soon; notify the maintenance lead.
Critical	Signs of an immediate hazard or imminent loss of function.	Follow emergency procedures; stop and isolate through approved steps; involve qualified personnel.

5.3 What to include when escalating

- Machine identifier, location, and operating mode at the time.
- Exact observation, time first noticed, and whether it is getting worse.
- Sensor values, trend screenshots or exports, alarm codes, and units.
- Recent maintenance, tool changes, process changes, and environmental conditions.
- Any model output (classification, score, threshold, input values) and the model version.
- What has already been checked, by whom, and what was ruled out.
- Immediate risk assessment and any actions taken to make the area safe.

6. Data quality, sensor validation, and baselines

Predictive maintenance is only as good as the data under it. Many apparent machine problems turn out to be data problems, and many real problems are hidden by bad data.

6.1 Sensor and data plausibility checks

Check	What to look for	Why it matters
Range	Values outside what is physically possible for the sensor or the machine.	Catches wiring faults and scaling errors.
Units	Mixed Kelvin/Celsius/Fahrenheit, rpm versus rad/s, Nm versus other units.	Unit mismatches create false alarms or hide problems.
Flatline	A signal that does not change at all across time.	Often a stuck sensor or a lost connection.
Spikes	Isolated extreme values that vanish at the next sample.	May be electrical noise rather than events.
Drift	Slow, steady offset relative to a trusted reference.	Calibration issues can mimic degradation.
Timestamps	Gaps, duplicates, clock offsets between systems.	Correlating signals needs aligned time.
Missing values	Dropouts and how they are filled.	Careless filling can create fake patterns.
Cross-check	Agreement between related signals (for example current versus torque).	Disagreement suggests a sensor fault.
Calibration status	Date of last calibration and traceability.	Uncalibrated data weakens every conclusion.

6.2 Building a valid baseline

- Define operating modes (idle, warm-up, steady load, heavy load, product types) and baseline each separately.
- Collect enough data across normal variation: shifts, ambient range, product mix.
- Exclude periods of known faults, maintenance, or abnormal operation.
- Record how the baseline was built (dates, filters, sample size) so it can be audited.
- Review baselines after major changes: tooling, process, repairs, relocation.
- Be careful with automatically rolling baselines: slow degradation can be absorbed into the 'normal' and go unnoticed.

6.3 Approved operating ranges versus baselines

A baseline describes what is typical for this machine. An approved operating range describes what the manufacturer or site engineering allows. They are different things. A machine can sit comfortably inside its baseline while drifting toward a limit, or can differ from the baseline while remaining within range. This guide supplies neither; obtain approved ranges from the OEM or site engineering and store them as separate, versioned sources.

7. The AI4I 2020 dataset: what it is and how to treat it

The AI4I 2020 Predictive Maintenance Dataset (S. Matzka, 2020; available from the UCI Machine Learning Repository) is a **synthetic** dataset designed to resemble real predictive-maintenance data. It is useful for learning, prototyping, and pipeline testing, but it is not measurements from a real factory.

7.1 Structure

The dataset has 10,000 records. Each row is one observation of a simulated machine. Columns, as described in the dataset documentation, include:

Field	Meaning	Notes
UDI	Row identifier (1 to 10,000).	An identifier only; exclude from model features.
Product ID	Letter L, M, or H plus a serial number.	Letter shows the quality variant; do not use the serial as a feature.
Type	Quality variant: L (low), M (medium), H (high).	About 50 percent L, 30 percent M, 20 percent H of products.
Air temperature	Ambient temperature in Kelvin.	Generated around 300 K with random variation.
Process temperature	Process temperature in Kelvin.	Generated as air temperature plus roughly 10 K with variation.
Rotational speed	Speed in rpm.	Derived from a power level with added noise.
Torque	Torque in Nm.	Roughly normally distributed around 40 Nm.
Tool wear	Accumulated wear in minutes.	Quality variants add different amounts of wear per cycle.
Machine failure	1 if any failure mode occurred, else 0.	The usual classification target.
TWF, HDF, PWF, OSF, RNF	Individual failure-mode flags.	Do not use as features when predicting Machine failure (see 7.3).

7.2 Failure modes in the dataset

The dataset documentation describes five failure modes, each produced by a rule in the data generator. These rules are properties of the synthetic data, **not engineering limits for any real machine**.

Code	Name	Dataset-generation idea (synthetic)
TWF	Tool wear failure	The tool fails or is replaced at a random point once wear reaches a high range.
HDF	Heat dissipation failure	Occurs when the temperature difference between process and air is small while rotational speed is low.
PWF	Power failure	Occurs when power (torque times angular speed) is outside a defined band, too low or too high.
OSF	Overstrain failure	Occurs when the product of tool wear and torque exceeds a level that depends on the quality variant.
RNF	Random failures	A very small chance of failure regardless of the parameters.

Verify and caveat
Check the exact generation rules and numbers against the dataset's official documentation before citing them in your project report. Whatever the rules are, they describe a simulation. A real machine's failure behaviour, thresholds, and sensor relationships will differ.

7.3 Practical modelling notes

- **Class imbalance.** Failures are a small minority (a few percent of rows). Accuracy alone is misleading: a model that always predicts 'no failure' scores high accuracy and catches nothing.
- **Target leakage.** The failure-mode columns (TWF, HDF, PWF, OSF, RNF) are components of the target. Using them as features inflates results without learning anything useful. Drop them, along with UDI and Product ID, for the main model.
- **Engineered features.** Temperature difference, mechanical power (torque x angular speed), and wear x torque are physically meaningful combinations. They can help models and also make explanations easier, but note they mirror the synthetic rules.
- **Splits.** Use stratified train/validation/test splits so each contains failures. Keep a held-out test set untouched until the end.
- **Units.** Temperatures are in Kelvin in the dataset. Display conversions (for example to Celsius) must not change what the model receives.
- **Generalisation.** Strong results on a synthetic dataset do not imply strong results on real equipment.

8. How to interpret an ML failure score

A machine-learning system may return a binary classification and a probability-like score. The classification depends on the model's decision threshold; the score is not a guarantee that a failure will or will not occur. A 'No failure' classification can still have a non-zero failure score if it is below the selected threshold.

8.1 Score, threshold, and classification

- The **score** is the model's estimated likelihood-like value for the failure class. Some models produce scores that are not well calibrated, so a score of 0.30 may not mean a 30 percent real-world chance.
- The **threshold** converts the score to a decision (for example, flag when the score is at or above the threshold). Lower thresholds catch more failures but raise false alarms; higher thresholds do the opposite.
- The **classification** (Failure / No failure) is only the result of that threshold. Report the score and the threshold together.

8.2 Reading the four outcomes

	Failure really occurs	No failure occurs
Model flags failure	True positive: a useful alert.	False positive: a false alarm (cost: wasted inspection, alert fatigue).
Model does not flag	False negative: a missed failure (cost: possible downtime or damage).	True negative: correct quiet.

In maintenance, the costs of false alarms and missed failures are rarely equal. The right threshold depends on failure consequences, inspection cost, and how many alerts people can realistically act on.

8.3 How the assistant should talk about a score

Do	Do not
Say the output is a prototype decision-support signal trained on synthetic data.	Call it a diagnosis or a certified prediction.
Report score and threshold together, and mention that a low score is not a guarantee.	Say 'the machine is safe' because the classification was 'No failure'.
Suggest a structured review (Sections 4 and 5) and an authorized inspection when concerns exist.	Tell anyone to keep operating equipment that shows signs of an immediate hazard.
Point out which inputs most influenced the score when explanation data is available, with caution.	Present feature importance as proof of a root cause.
Encourage recording the outcome so the model can be evaluated later.	Invent machine-specific limits or repair steps.

8.4 Explanations and their limits

Feature-importance or per-prediction explanations (for example SHAP-style attributions) show which inputs the model relied on, not why the machine is behaving as it is. On the synthetic dataset, important features often reflect how the data was generated. Present explanations as 'what the model used', not as engineering findings.

8.5 Monitoring a model over time

- Track score distributions, alert rates, and confirmed outcomes (precision, recall) over time.

- Watch for input drift: new products, sensors, tools, or operating regimes that differ from training data.
- Record model version, training data version, threshold, and evaluation date with every deployment.
- Define who may change thresholds and how changes are approved and documented.

9. Evaluating the model responsibly

The current model is trained on the AI4I 2020 synthetic dataset. Treat its output as a prototype decision-support signal, not a certified diagnosis or an instruction to keep operating unsafe equipment. Evaluate precision, recall, false alarms, and missed failures on held-out data, and validate with relevant real-world data before operational use.

9.1 Metrics

Metric	Question it answers	Caution
Precision	Of the alerts raised, how many were real failures?	Can look high if the model flags very few cases.
Recall (sensitivity)	Of the real failures, how many were caught?	Can be pushed up by flagging everything.
F1 / F-beta	A single balance of precision and recall (beta above 1 favours recall).	Hides the trade-off; always show both parts.
Confusion matrix	Exact counts of TP, FP, FN, TN.	Counts matter more than percentages with rare failures.
PR curve / average precision	Performance across thresholds for the rare class.	More informative than ROC with heavy imbalance.
ROC-AUC	Ranking ability across thresholds.	Can look optimistic when negatives dominate.
Calibration	Do scores behave like probabilities?	Needed if scores are shown as percentages.
Accuracy	Overall fraction correct.	Misleading with rare failures; never use alone.

9.2 Good practice checklist

	Item
[ ]	Use stratified splits and keep a final test set untouched until the end.
[ ]	Remove leakage columns (failure-mode flags, identifiers) from the main model's features.
[ ]	Compare with simple baselines (for example a rule-based or logistic model) before choosing a complex one.
[ ]	Choose the threshold on validation data, using the cost of false alarms versus misses; then report test results once.
[ ]	Report precision, recall, confusion matrix, and PR curve, not accuracy alone.
[ ]	Analyse errors by machine type (L/M/H) and by failure mode where labels allow.
[ ]	Run robustness checks: noisy inputs, missing values, unit mistakes, out-of-range inputs.
[ ]	State clearly that the data is synthetic and what real-world validation is still required.
[ ]	Document the model: purpose, data, features, version, metrics, threshold, known limits.

9.3 Moving toward real-world use

- Collect labelled real data: confirmed failures, near-misses, maintenance actions, and normal operation.
- Run the model in shadow mode (no operational decisions) and compare its alerts with outcomes.

- Involve maintenance engineers in setting thresholds and alert wording.
- Define a human review step for every alert before any action.
- Re-evaluate after process, tooling, or sensor changes.

10. Records, checklists, and templates

Consistent records make troubleshooting faster, support audits, and provide the labelled history that predictive models need.

10.1 Preventive maintenance record checklist

	Item
[ ]	Record machine identifier, model, and operating context.
[ ]	Record date/time and operator according to site policy.
[ ]	Capture sensor readings and alarm codes.
[ ]	Note deviations from baseline and measurement conditions.
[ ]	Document inspection findings and maintenance actions.
[ ]	Confirm parts, lubricants, and service intervals against OEM documentation.
[ ]	Follow safety isolation and authorization requirements.
[ ]	Complete and review post-maintenance checks.

10.2 Operator observation checklist (start of shift or round)

For observations that can be made from a safe position without removing guards or entering hazardous zones.

	Item
[ ]	Guards, interlocks, and emergency stops appear in place and undamaged (report issues; do not bypass).
[ ]	Any new or unusual noise, vibration, smell, or visible heat noted.
[ ]	Visible leaks, residue, or debris around the machine noted.
[ ]	Displayed readings (temperature, speed, torque/load) look plausible and stable for the current mode.
[ ]	Active alarms or warnings noted with codes and time.
[ ]	Output quality changes noted (finish, dimensions, defects).
[ ]	Tool changes and counter resets since last shift confirmed.
[ ]	Handover notes read and any open issues acknowledged.

10.3 Post-maintenance verification checklist

	Item
[ ]	Work completed and documented by an authorized person.
[ ]	Isolation removed only through the site-approved procedure; area clear of people and tools.
[ ]	Guards, covers, and interlocks restored and functioning.
[ ]	Start-up and initial running observed using approved procedures.
[ ]	Key readings compared with the pre-fault baseline for the same mode.
[ ]	Tool counters and maintenance records updated.

	Item
[ ]	Follow-up monitoring period defined and communicated.
[ ]	Findings and parts used recorded for future reference.

10.4 Observation log template

Field	Example entry (fictional)
Date / time	2026-10-30 14:20
Machine ID / mode	Line-3 spindle / steady load, product type M
Observation	Process temperature trending above the usual level for this mode over the past three shifts.
Readings (with units)	Air temp, process temp, speed (rpm), torque (Nm), tool wear (min), alarm codes.
Context	Warm day; filter last cleaned two weeks ago; no tool change this shift.
Checks done	Sensor cross-checked with second reading; visual check from safe position.
Risk assessment	Watch level; no immediate hazard observed.
Action / escalation	Reported to maintenance lead; inspection requested; monitoring frequency increased.
Outcome (added later)	Cause identified and corrected by authorized technician; verified normal.

11. Safety fundamentals

Read first
This section is general awareness only. It is not a lockout/tagout procedure, a risk assessment, or authorization to perform any task. Follow your site's approved procedures and the manufacturer's instructions, and use qualified personnel.

11.1 Principles

- **Safety before diagnosis.** A suspected hazard always comes before data analysis.
- **Authorization.** Only trained, authorized people may open guards, perform electrical work, work on pressurised systems, or carry out repairs.
- **Energy isolation.** Machines can hold hazardous energy: electrical, mechanical (moving or stored), hydraulic, pneumatic, thermal, gravitational, chemical. Isolation and verification are done only through site-approved procedures such as lockout/tagout.
- **Never bypass guards or interlocks**, even briefly, even to 'see what's happening'.
- **Hot, rotating, and pressurised parts** can injure without warning. Stay clear unless the task and procedure allow otherwise.
- **Protective equipment** must match the site's hazard assessment and the manufacturer's instructions.
- **Chemicals and lubricants** are handled according to their safety data sheets and site rules.
- **Stop-work authority.** Anyone may raise a concern; if in doubt, stop and ask.

11.2 Signs of an immediate hazard

- Smoke, flames, sparks, or a burning smell.
- Rapid or runaway temperature rise.
- Severe or sudden vibration, loud bangs, or grinding.
- Fluid or gas release, spraying, or pooling near energized equipment.
- Damaged, missing, or bypassed guards and interlocks; emergency stop not working.
- Any situation where people could be injured if the machine continues.

If any of these occur, follow site emergency procedures, alert qualified personnel, and keep people away. Do not rely on a model score or on this guide to decide that it is safe to continue.

11.3 Responsibilities

Role	Typical responsibility
Operator	Observe from a safe position; report abnormal conditions; follow procedures; use stop-work authority.
Maintenance technician	Perform authorized inspection and repair using OEM and site procedures, including isolation and verification.
Maintenance engineer / reliability	Define baselines, limits, strategies, and escalation rules; review model outputs and outcomes.
Safety officer	Maintain safety procedures; advise on risk; investigate incidents.
Data / ML owner	Maintain data quality, model versions, evaluation, and documentation; do not present outputs beyond their validated scope.

12. Worked scenarios (general, fictional)

Each scenario shows a reasoning pattern, not a prescribed repair. Numbers are deliberately omitted because limits depend on the specific machine.

Scenario 1: Process temperature creeping up over several weeks

- **Observation:** Gradual rise versus the baseline for the same mode; torque and speed look normal.
- **First checks:** Units and sensor validity; ambient trend; recent changes in ventilation or cleaning schedule.
- **Possible hypotheses:** Reduced heat removal; sensor drift; slow increase in friction.
- **Avoid concluding:** That a specific component is failing based on temperature alone.
- **Next:** Raise to Watch; schedule an authorized inspection of cooling and related items per OEM guidance; keep trending.

Scenario 2: Torque spikes at the start of each shift

- **Observation:** Short torque peaks during start-up that are absent later.
- **First checks:** Is the machine at operating temperature? Is the warm-up routine followed? Any change in material or setup?
- **Possible hypotheses:** Cold-start friction; start-up sequence differences; process engagement issues.
- **Avoid concluding:** That these are faults before comparing against start-up baselines for the same mode.
- **Next:** Create a separate start-up baseline; escalate if peaks grow or are accompanied by noise or alarms.

Scenario 3: Tool wear counter high, model score rising

- **Observation:** Wear counter near the site's inspection point; torque slightly higher; the model's failure score trending up.
- **First checks:** Was the counter reset at the last change? Output quality trend? Torque versus similar earlier runs.
- **Possible hypotheses:** Tool nearing end of life; rising load from other causes.
- **Avoid concluding:** That the model has 'predicted' the date of failure.
- **Next:** Plan tool inspection or replacement by the approved method and schedule; record what is found.

Scenario 4: Model says 'No failure' but the operator hears new noise

- **Observation:** Low score below threshold; operator reports new grinding or knocking.
- **First checks:** Safety triage (is there an immediate hazard?); vibration or temperature changes if measured.
- **Interpretation:** The model has no acoustic input and was trained on synthetic data. A quiet score does not override a credible observation.
- **Next:** Escalate for an authorized inspection; log the case as a potential missed alert for future model review.

Scenario 5: High failure score but everything looks normal

- **Observation:** Score at or above threshold; readings look ordinary to the operator.
- **First checks:** Input values and units fed to the model; sensor plausibility; recent distribution shift (new product type, new tool).

- **Possible explanations:** Combined subtle shift; data error; model sensitivity or false alarm.
- **Next:** Compare with baseline, have an authorized person check the machine, and record the outcome as feedback.

Scenario 6: Several signals go flat at once

- **Observation:** Temperature, speed, and torque stay constant to many decimals for a long period.
- **First checks:** Data logger, network, and power; timestamps; whether the machine is actually running.
- **Interpretation:** More consistent with a data pipeline fault than with ideal machine behaviour.
- **Next:** Repair the data path through the proper channel; do not use the flat data for model decisions.

13. Building and governing the RAG knowledge base

A Retrieval-Augmented Generation assistant answers by retrieving passages from documents and using them to write a response. Its quality depends on the documents, how they are split and labelled, and the rules that govern answers.

13.1 Document metadata

Keep document titles, revision dates, source URLs, and applicability metadata so the assistant can cite the right source and avoid mixing instructions for different equipment. A suggested schema:

Field	Purpose	Example
doc_id	Stable unique identifier.	mechmind-general-guide
title	Human-readable title used in citations.	General Predictive Maintenance & Troubleshooting Guide
version / revision_date	Detect outdated guidance.	2.0 / 2026-10-30
source_type	general_guide, oem_manual, site_procedure, dataset_doc, model_card.	general_guide
applicability	Which machines, models, or sites it applies to.	General; not machine-specific
authority_level	Order of precedence when sources conflict.	background
section / page	Locate the passage for citation.	Section 8.3, page 14
source_url / owner	Where it came from and who maintains it.	Internal repository / maintenance team
safety_flag	Marks passages that must be shown with safety wording.	true

13.2 Chunking and retrieval tips

- Split on headings and keep each chunk self-contained: include the section title and, for tables, the header row, so a chunk makes sense alone.
- Aim for focused chunks (roughly a few paragraphs). Avoid cutting through a table or a numbered procedure.
- Add the document title and version to each chunk's metadata, not only to the first one.
- Keep OEM manuals separate from this general guide and filter by machine model so instructions are never mixed.
- Prefer hybrid retrieval (keyword plus semantic) because part numbers, alarm codes, and abbreviations are exact-match terms.
- Re-index whenever a document changes, and keep old versions out of the active index (or clearly marked).

13.3 Answering rules for the assistant

- **Ground answers in retrieved text** and cite the document and section.
- **Say when the knowledge base does not contain the answer** instead of guessing, especially for limits, torque values, lubricants, or repair steps.
- **Safety first.** If the question describes an immediate hazard, lead with site emergency procedures and qualified personnel.

- **Separate general guidance from machine-specific facts.** If no OEM document for the machine is present, say so.
- **Describe model outputs correctly** as prototype signals with scores and thresholds (Section 8).
- **Do not provide lockout/tagout steps or repair instructions** from this general guide; refer to approved procedures.
- **Ask for missing context** (machine, mode, units, values) when it materially changes the answer.

13.4 Example out-of-scope requests and suitable responses

User request	Suitable response
'What is the maximum safe temperature for my spindle?'	Explain that this depends on the machine; point to the OEM documentation and approved operating range; offer to explain how to compare with a baseline.
'Give me the torque value to tighten the bearing cap.'	Decline to supply a specific value; refer to the OEM manual and authorized technician; mention that guessing can cause damage or injury.
'How do I lock out this machine?'	Explain that lockout/tagout must follow the site's approved procedure; offer general concepts and safety principles only.
'The model says no failure, so can I keep running it with the burning smell?'	Say no: a burning smell is an immediate-hazard sign; follow emergency procedures; a model score does not override it.

13.5 Evaluating the assistant

- Build a test set of questions with expected source sections (for example the questions in Section 14).
- Measure retrieval quality (did the right chunk appear?) separately from answer quality (was the answer correct, cited, and safe?).
- Include adversarial tests: requests for machine-specific limits, requests to ignore safety, prompts that mix machine types.
- Review failures, then fix the documents, metadata, chunking, or prompts, in that order of preference.
- Re-run the test set after every document or model change.

13.6 Growing the knowledge base

- Add OEM manuals, site procedures, maintenance histories, and model cards, each with proper metadata.
- Add a short 'applicability' statement at the top of each new document.
- Retire or mark superseded documents instead of leaving conflicting versions active.
- Log which documents were retrieved for each answer to support audits and improvement.

14. Frequently asked questions

Q. Which indicators should be reviewed when temperature rises unexpectedly?

Review torque/load, rotational speed, tool wear, vibration and noise, alarms, ambient temperature, cooling and ventilation, and recent maintenance or process changes. Check units and sensor validity first, then compare with a baseline for the same operating mode.

Q. Why should readings be compared with a baseline from similar operating conditions?

Readings change naturally with mode, load, product, and ambient conditions. Comparing across different conditions can create false alarms or hide real changes.

Q. What should be recorded before escalating a suspected machine issue?

Machine identifier, time, operating mode, sensor values with units, alarm codes, recent changes, what has been checked, any model output, and an immediate risk assessment (Section 5.3).

Q. Why is a model's failure probability not the same as a confirmed failure?

It is an estimate from a model trained on data, converted to a decision by a threshold. It can produce false alarms and misses, and the current model is trained on synthetic data.

Q. What should an operator do if there are signs of an immediate safety hazard?

Follow site emergency procedures, keep people away, and involve qualified personnel. Do not bypass guards or interlocks, and do not wait for data analysis.

Q. Can this guide tell me the allowed temperature, speed, or torque for my machine?

No. These limits are machine-specific and must come from the manufacturer's documentation or site engineering.

Q. Does a 'No failure' result mean the machine is safe?

No. It means the score was below the model's threshold. It is not a guarantee and does not replace inspection or safety procedures.

Q. What does a high tool-wear value mean?

It means accumulated usage is high relative to the counter's scale. Compare against the approved inspection or replacement schedule, confirm the counter was reset at the last tool change, and inspect using approved methods.

Q. Why is accuracy a poor metric for this dataset?

Failures are rare. A model that predicts 'no failure' for every case would score high accuracy but would catch none. Use precision, recall, the confusion matrix, and the PR curve.

Q. Why remove TWF, HDF, PWF, OSF, and RNF from the features?

They are components of the target (Machine failure). Including them leaks the answer and inflates results.

Q. Are the AI4I 2020 patterns valid thresholds for real equipment?

No. The data is synthetic. Use it for learning and prototyping, and validate with real data before operational use.

Q. What if the vibration or noise seems severe?

Treat it as a potential immediate hazard: follow emergency procedures and involve qualified personnel.

Q. How often should baselines be updated?

After significant changes (tooling, process, repairs, relocation) and on a defined review schedule. Avoid automatic updates that could absorb slow degradation.

Q. Can I use this guide as a lockout/tagout procedure?

No. Lockout/tagout must follow your site's approved procedure and be performed by authorized people.

Q. What should I add next to make the assistant more useful?

OEM manuals for your specific machines, site procedures, maintenance histories, alarm-code lists, and a model card, each with version and applicability metadata.

15. Glossary

Term	Meaning
Alarm	A machine or control notification that a condition needs attention.
Baseline	Typical readings for a machine in a defined operating mode, used for comparison.
Calibration	Checking and adjusting a sensor against a reference so readings are accurate.
Class imbalance	A situation where one class (such as failure) is much rarer than the other.
Concept drift	A change over time in how inputs relate to outcomes, which can degrade a model.
Condition-based maintenance	Maintenance triggered by measured condition rather than by the calendar.
Confusion matrix	A table of true/false positives and negatives for a classifier.
Data leakage	When information unavailable at prediction time (or the target itself) enters the features.
Decision threshold	The score level at or above which the model flags a failure.
False negative	A real failure the model did not flag.
False positive	A flag raised when no failure occurred.
Failure mode	The way in which something can fail (for example overheating, wear, overload).
Functional failure (F)	The point at which equipment can no longer perform its function.
Hybrid retrieval	Combining keyword and semantic (vector) search.
Interlock	A safety device that prevents operation when a guard is open or a condition is unsafe.
Lockout/tagout (LOTO)	A site procedure to isolate and secure hazardous energy before work. Follow the approved procedure.
OEM	Original equipment manufacturer.
P-F interval	The time between first detectable change and functional failure.
Precision	Proportion of flagged cases that were real failures.
Predictive maintenance	Using data and models to anticipate developing problems and plan action.
RAG	Retrieval-Augmented Generation: answering with text retrieved from a knowledge base.
Recall	Proportion of real failures that were flagged.
Remaining useful life (RUL)	An estimate of how long equipment can continue to function.
Root cause	The underlying reason a problem occurred, as opposed to its symptoms.
Setpoint	The target value a control system tries to achieve.
Synthetic data	Artificially generated data that imitates real data.
Tool wear	Accumulated usage or degradation of a cutting or working tool.
Torque	Rotational force, commonly measured in newton-metres (Nm).
Vibration signature	The characteristic pattern of vibration of a machine in normal operation.

16. References, limitations, and revision log

16.1 Suggested references (verify current editions)

- S. Matzka, 'Explainable Artificial Intelligence for Predictive Maintenance Applications', 2020 Third International Conference on Artificial Intelligence for Industries (AI4I); dataset: AI4I 2020 Predictive Maintenance Dataset, UCI Machine Learning Repository.
- ISO 17359: Condition monitoring and diagnostics of machines, general guidelines.
- ISO 13374: Condition monitoring and diagnostics of machines, data processing, communication and presentation.
- ISO 20816 series: Mechanical vibration, measurement and evaluation of machine vibration.
- Applicable lockout/tagout and machine-safeguarding regulations and standards for your jurisdiction and industry.
- The OEM documentation for each machine in your facility.

16.2 Limitations

This guide is intentionally general. It does not identify a particular machine, establish permissible operating limits, diagnose a specific fault, or replace a qualified technician. It is not legal, regulatory, or safety certification advice. Add OEM-approved manuals and site-approved procedures when available. Keep document titles, revision dates, source URLs, and applicability metadata so the RAG assistant can cite the right source and avoid mixing instructions for different equipment.

16.3 Revision log

Version	Date	Summary of changes
1.0	Initial	Starter guide: purpose, indicators, workflow, ML score note, PM checklist, limitations.
2.0	2026-10-30	Expanded edition: maintenance strategies, detailed indicators, indicator patterns, severity levels, data quality, AI4I 2020 dataset notes, model evaluation, checklists and templates, safety fundamentals, scenarios, RAG governance, FAQ, glossary, references.

End of guide - MechMind AI prototype knowledge base